ORION Paper III: A Recoverable Global Knowledge Portrait from Source-Local Scientific Projections

Working framework draft

August 2026

Abstract

Cross-domain scientific synthesis is vulnerable to false integration: identical names may denote different referents, nominally similar constructs may use incompatible operationalizations, and apparently contradictory claims may differ in context, modality or attribution. We present a recoverable global knowledge portrait built from source-local scientific projections. Before global integration, proposed mappings distinguish referent identity, construct identity, measurement/operationalization, context and representation equivalence. Compatible projections may be GLUEd under explicit preservation conditions; incompatible or unresolved views remain typed OBSTRUCTION or PLURAL_VIEW charts rather than being forced into one canonical summary. Local exact worlds falsify the mapping semantics. In a disjoint, prospectively execution-frozen 32-case confirmatory holdout assembled from pinned external human/expert resources, the ORION mapping calculus made zero false merges versus 0.1875 for flat predicate canonicalization; the paired difference was -0.1875 (95% bootstrap CI $[-0.34375, -0.0625]$). Against an exact-coordinate conservative control, the false-split difference was 0.000 with interval $[0.000, 0.000]$, satisfying the predeclared confirmatory rule. This result concerns already-structured scientific projections only. End-to-end adequacy from raw scientific text, recoverability in generated portraits, downstream utility, and the broader eight-family cross-domain protocol remain `CANNOT_CHECK`.

Keywords: scientific knowledge integration; provenance; schema alignment; measurement harmonization; scientific pluralism; knowledge representation.

1 Introduction

Scientific claims are distributed across disciplines, each with its own terminology, measurement practices, contexts, and representational choices. A global synthesis can therefore fail even when every local statement is individually well formed: identical surface terms can refer to different objects, different labels can denote compatible objects, and apparently contradictory claims can differ only in modality, attribution, or context. The central problem of this paper is not retrieval itself, but the authority to identify two already-structured scientific representations as one global statement.

1.1 The proposal in brief

ORION Paper III represents each source as a *source-local scientific projection* bound to its provenance. A proposed cross-source mapping records scientific coordinates such as referent, construct, measurement, context, polarity, modality, and attribution separately rather than reducing them to one similarity score. When the coordinates required by the claim are compatible, the projections may be GLUEd. When a load-bearing coordinate is incompatible or unresolved, the mapping remains a typed *obstruction* or plural view rather than being forced into a canonical merge.

This paper evaluates that mapping rule at the layer for which external evidence currently exists. The initial public-reference study contains 32 already-structured cases derived from pinned external sources. A second, disjoint 32-case holdout was selected and execution-frozen before its system outputs, including its exact gold hash, source revisions, evaluator code, bootstrap seed, margins, and pass rule. On that confirmatory holdout, ORION produced zero false merges versus 0.1875 for flat predicate canonicalization; the paired difference was -0.1875 with a 95% bootstrap interval of $[-0.34375, -0.0625]$. The false-split difference versus an exact-coordinate conservative control was 0.000 with interval $[0.000, 0.000]$, satisfying the predeclared confirmatory rule.

The evidence is intentionally narrower than the full Global Knowledge Portrait programme. It does not establish raw-text extraction superiority, expert eight-family construct validity, reader-level recoverability of a generated portrait, downstream scientific-answer improvement, or necessity of every semantic coordinate. Those are follow-up claims and remain `CANNOT_CHECK` rather than being inferred from the mapping result.

1.2 Claims and non-claims

The paper absorbs the following mechanism families as prior work rather than claiming them independently:

- source-grounded scientific structure extraction and knowledge assembly (MUSE [6], Discovery Engine [2]);
- multidisciplinary scientific schemas (SciSchema.org [7]);
- evidence-linked schema induction and fusion (SCOPE/SCION [11]);
- provenance-aware schema-conditioned integration and retrieval (Executable Schema Contracts [13]);
- scientific entity/relation extraction (SciER [20]);
- ontology and schema alignment (Ontology Matching [8], LLMATCH [19]);
- heterogeneous knowledge-graph construction, entity resolution, fusion, and provenance [10, 17];
- stance and claim-comparison methods [9];
- literature-based discovery [15, 16];
- scientific pluralism [14, 5];
- retrieval-augmented scientific synthesis and cross-domain agent orchestration [1, 18].

The surviving scoped empirical claim is P3.C5: for the frozen public-reference mapping task, explicit scientific-coordinate conditions plus an obstruction state reduce false merges relative to flat predicate canonicalization while satisfying the predeclared false-split guard. The initial 32 cases are not pooled into the confirmatory verdict. Zero-effect coordinate ablations are retained as limitations, not interpreted as proof that those coordinates are unnecessary.

1.3 Structure of the paper

Section 2 establishes the absorbed prior-work boundary. Section 3 defines source projections, mappings, GLUE, and obstruction. Section 4 describes the two public-reference mapping datasets and their provenance. Section 5 records the prospective confirmatory design and controls. Section 6 reports the frozen result and ablations. Section 7 separates the supported mapping claim from the broader unexecuted programme, and Section 8 states the bounded conclusion.

2 Related work

ORION Paper III's contribution is deliberately narrow. Rather than claiming novelty in extraction, schema matching, data integration, contradiction/stance analysis, or literature-based discovery, we treat those capabilities as prior work and ask a smaller question: whether coordinate-wise scientific meaning can govern when already-structured source projections may be merged and when a provenance-bound obstruction should remain visible.

2.1 Structured scientific knowledge bases and extraction

MUSE [6] constructs a full-text, multi-domain resource of source-grounded problem–solution–rationale structures. This absorbs source-grounded cross-domain scientific structure extraction as prior work. Its representational target is problem, solution, rationale, and their links; our scoped object instead tests whether referent, construct, measurement, context, modality, and attribution distinctions can govern a downstream mapping decision and produce an obstruction when compatibility is not earned. The Discovery Engine [2] likewise distils literature into structured knowledge artifacts and an integrated computational landscape. Both are important neighboring knowledge-assembly systems, not mechanisms that ORION may relabel as its own.

SciER [20] provides scientific entity and relation annotations for datasets, methods, and tasks in full-text scientific documents. Scientific information extraction, discourse processing, and entity/relation recovery are therefore outside the scoped novelty claim. The residual question is downstream of extraction: given structured projections, which coordinate differences license a merge and which require preservation as an obstruction.

2.2 Schema-centric approaches

SciSchema.org [7] provides multidisciplinary, expert-reviewed scientific-process schemas with parameters, measurements, procedures, and provenance. This is strong prior art for reusable scientific schema design. ORION does not claim expert schema construction; its scoped mapping result instead treats each source projection as locally addressable and asks whether a proposed cross-source mapping preserves the coordinates required for the claim at hand.

SCOPE and SCION [11] introduce an auditable benchmark and reference pipeline for evidence-linked schema induction and optional fusion from text. Schema induction, evidence-linked candidate construction, and conservative fusion are therefore absorbed. The P3 discriminator is different: it evaluates a coordinate-wise GLUE-or-obstruction rule over already-structured scientific projections, including cases in which refusing a merge is the correct outcome.

Executable Schema Contracts [13] automatically discover an executable schema from heterogeneous sources and use it to drive provenance-aware construction and multi-source retrieval. This absorbs schema-conditioned integration, provenance-aware construction, and retrieval. ORION's bounded residual is not shared-schema discovery; it is explicit retention of the conditions and non-preserved scientific coordinates that can make a proposed mapping inadmissible.

2.3 Alignment, integration, and resolution

Ontology matching [8] explicitly studies correspondences such as equivalence, subsumption, and disjointness between ontology entities. LLM-based systems such as LLMATCH [19] add modern schema-preparation, candidate-selection, and column-alignment machinery. These mechanisms are prior art. P3's scoped distinction is to bind a mapping decision to several scientific coordinates simultaneously and retain exact source/provenance and non-preserved-coordinate information for the resulting GLUE or obstruction decision.

Modern knowledge-graph construction surveys [10] cover heterogeneous-source integration, ontology/schema matching, entity resolution, fusion, provenance, and quality assurance, while reviews of scholarly knowledge graphs [17] cover extraction, construction, refinement, and use of semantic scholarly infrastructures. Those families are not P3 novelty. Entity resolution and fusion are especially useful pressure cases because a record-level match may be appropriate even when a scientific claim-level merge remains unsafe due to construct, measurement, context, or modality differences.

2.4 Claim comparison and stance

Stance detection assigns target-relative labels such as support/opposition or other task-specific stance categories; surveys document its use in fact-checking and misinformation settings [9]. P3 does not claim stance detection. Its scoped claim is that apparent scientific disagreement may arise from different context, modality, measurement, or attribution coordinates, and that a mapping system should preserve those reasons rather than convert every difference into one generic opposition or contradiction label.

2.5 Measurement equivalence and construct validity

Measurement-equivalence and replication literatures have long established that nominally identical outcome measures need not be comparable across settings and that equivalence can require explicit testing [12]. ORION does not claim this principle. It makes measurement/operationalization one explicit coordinate in the scientific mapping record and allows an unresolved measurement relation to block a global merge.

2.6 Literature-based discovery

Swanson's literature-based discovery lineage [15, 16] shows how complementary literatures can expose implicit connections without direct citation links. Discovery of cross-literature bridges is therefore prior art. P3 adds only a bounded mapping requirement: the bridge concept must retain compatible scientific meaning across the two source-local projections; otherwise the bridge remains an obstruction rather than being promoted as a scientific integration.

2.7 Scientific pluralism

Scientific pluralism has long argued that multiple representations may remain legitimate without collapsing into one complete account [14, 5]. ORION does not claim pluralism as a new philosophical result. The computational question is whether an integration record can preserve a plural or obstructed outcome, with exact source lineage, instead of forcing a canonical merged statement.

Mechanism family	Absorbed prior work	Scoped P3 residual
Source-grounded structured KB	MUSE, Discovery Engine	Coordinate-governed mapping of source-local projections
Expert scientific schemas	SciSchema.org	Mapping may remain obstructed rather than globally canonical
Schema induction/fusion	SCOPE/SCION	Conditional GLUE with explicit obstruction
Provenance-aware integration	Executable Schema Contracts	Preserve non-preserved scientific coordinates
Scientific IE	SciER	Mapping semantics downstream of extraction
Ontology/schema alignment	Ontology Matching, LLMATCH	Scientific-coordinate conditions on mapping
Measurement equivalence	Measurement-invariance literature	Measurement as an explicit blocking coordinate
KG integration / entity fusion	KG-construction literature	Claim-level non-merge remains possible
Stance / claim comparison	Stance-detection literature	Typed reason for apparent disagreement
Literature-based discovery	Swanson lineage	Meaning-stability requirement on bridges
Scientific pluralism	Philosophy of science	Provenance-bound plural/obstructed output
Scientific RAG / cross-domain synthesis	OpenScholar, BioSage	Evaluate mapping rule independently of retrieval/synthesis

Table 1: The absorbed boundary for the scoped paper. The middle column is prior work; the last column states the narrower object tested by P3.C5.

2.8 Scientific RAG and cross-domain synthesis

OpenScholar [1] provides retrieval-augmented scientific literature synthesis at scale, while BioSage [18] combines retrieval, specialized agents, cross-disciplinary translation, and reasoning for scientific tasks. Retrieval, synthesis, agent orchestration, and cross-domain translation are therefore baseline capabilities, not P3 novelty. Likewise, LLM-assisted schema matching and induction are represented by LLMATCH and SCOPE/SCION [19, 11]. These systems address important acquisition and synthesis problems; the scoped P3 experiment asks a separate construct-validity question about the decision rule used after scientific projections are already structured.

2.9 Summary of the absorbed boundary and residual

The current empirical claim is not that this combination solves raw-text scientific integration end to end. It is the narrower P3.C5 result: on a disjoint, prospectively frozen public-reference holdout of already-structured scientific projections, the typed mapping calculus reduced false merges versus flat predicate canonicalization while satisfying the predeclared false-split guard. Raw-text extraction, full expert-atlas construct validity, recoverability, and downstream utility remain separate follow-up claims.

3 Method

This section defines the scientific mapping objects used by the scoped public-reference evaluation. Their original broader design lineage is preserved in `P3.cross-domain-atlas.v1`; the executed mapping protocol is the additive `P3.public-reference-mapping.v1.1-confirmatory`. The definitions are data-level and do not depend on a particular language model or retrieval backend.

3.1 Source and source-local projection

Let a source record s bind an external document or structured source item to an exact identity, version/revision, locator, and content commitment. ORION represents the relevant scientific meaning as a source-local projection

$$P_s = \langle R_s, C_s, M_s, X_s, \rho_s, \mu_s, a_s, \delta_s, \Pi_s \rangle, \tag{1}$$

with coordinates for referent R_s, construct C_s, measurement/operationalization M_s, context X_s, polarity ρ_s, modality μ_s, attribution a_s, discourse relation δ_s, and assumptions/unresolved information Π_s.

Each coordinate is a typed hypothesis supported by the source authority available for that case, not a value filled merely to complete a schema. In the scoped public-reference datasets, unsupported coordinates remain absent or unresolved; the study does not ask a model to infer missing gold coordinates from raw papers.

3.2 Coordinate-wise comparison

A comparison between projections P_a and P_b is a vector of typed coordinate relations rather than a scalar similarity score:

$$\begin{aligned} q(P_a, P_b) = \langle &\mathrm{ref}(R_a, R_b), \mathrm{const}(C_a, C_b), \mathrm{meas}(M_a, M_b), \mathrm{ctx}(X_a, X_b), \\ &\mathrm{pol}(\rho_a, \rho_b), \mathrm{mod}(\mu_a, \mu_b), \mathrm{attr}(a_a, a_b), \mathrm{disc}(\delta_a, \delta_b) \rangle. \end{aligned} \tag{2}$$

Relations can express compatibility, incompatibility, conditional transformability, or unresolved evidence according to the frozen semantic taxonomies. A proposed relation does not acquire authority merely from lexical or embedding similarity.

3.3 Typed mapping and preservation conditions

A typed mapping m states which coordinate relations it licenses and under what preservation conditions. Its record contains:

- the proposed relation (for example identity, equivalence, conditional transform, subsumption, association-only, no-licensed-mapping, or unresolved);
- the coordinates the mapping claims to preserve;
- conditions that must hold for the mapping to be used; and
- coordinates or information that are not preserved and therefore may not be silently transferred into a global statement.

This representation is intentionally stricter than a successful match score: a mapping can be structurally plausible yet inadmissible for the scientific claim if a load-bearing condition is absent.

3.4 GLUE, obstruction, and unresolved mapping

Two or more projections may be GLUEd only when the coordinate relations required by the mapping are compatible and its preservation conditions are satisfied. If a required relation is incompatible, the correct output is a typed OBSTRUCTION naming the failing coordinate or condition. If the available evidence is insufficient, the relation remains UNRESOLVED rather than being forced to merge or split. Multiple locally valid but mutually incompatible views may remain a PLURAL_VIEW.

The scoped confirmatory experiment tests this decision semantics directly. Its strongest discrimination is on polarity/modality/attribution/context cases, where flat predicate canonicalization merges claims that the typed rule keeps separate. It does not identify the necessity of every coordinate; coordinate necessity requires targeted cases in follow-up issue #280.

3.5 Provenance and recoverable records

Every scoped case retains the external source identity and the frozen record used to derive the expected mapping decision. A mapping result retains the source records and the coordinate conditions supporting its terminal. This provenance is necessary for independent replay of P3.C5.

The broader ORION design goes further and defines a global portrait

$$\mathcal{G} = \langle \{P_i\}, \mathcal{M}, \sigma \rangle, \tag{3}$$

where $\mathcal{M}$ stores accepted and rejected mappings and σ maps a global statement back to source projections and conditions. Reader-level recoverability of generated portraits is a follow-up empirical claim (P3.C8/ broader protocol), not a result of the current mapping study.

3.6 Search-universe feedback is outside the scoped empirical claim

The wider ORION architecture allows newly absorbed representations to change the search universe W and reopen future acquisition routes. That feedback is part of the broader programme but is not manipulated or measured in the public-reference mapping experiment. No result in this paper is attributed to a "W effect," and no W-expansion ablation is used to support P3.C5.

3.7 What is and is not claimed

ORION does not claim the first scientific knowledge graph, cross-domain RAG system, schema matcher, provenance graph, entity resolver, or literature-discovery method. The supported empirical object is narrower: over the frozen public-reference mapping cases, explicit scientific-coordinate conditions plus an obstruction state reduce false merges relative to flat predicate canonicalization while satisfying a conservative false-split guard.

Local exact worlds [3, 4] continue to falsify implementation semantics in controlled cases. They do not establish external raw-text extraction quality, recoverability, downstream utility, or broad cross-domain generality; those remain separate prospective questions.

3.8 Method-structure projection

Scientific meaning coordinates describe what a source claims. For downstream structural learning ORION also needs a source-local account of *how the source's method transforms its problem.* Paper III therefore adds the versioned `MethodStructureProjection.v1` object. It is derived from a provenance-bound Paper-I `MethodRealization.v1` and retains exact source/document/version/span identity, the target role and initial obstruction, assumptions/preconditions, representation changes and auxiliary objects where source-supported, the transformation/mechanic graph, invariants, progress and termination semantics, reconstruction to the original target, failure/negative-result statements, explicit author rationale when the source actually supplies it, and typed `UNKNOWN` coordinates otherwise.

This projection remains source-local. In particular, absence of an author rationale is not permission for ORION to invent one. The projection stores the exact realization and span digests that licensed each structured interpretation and has no authority to declare a Paper-VI method fibre.

3.9 Cross-domain method alignment and plural views

Given two source-local projections, Paper III may propose a claim-relative mapping μ by comparing declared coordinates such as preconditions, assumptions, invariants, progress, termination and reconstruction. The alignment state is one of `ALIGNED`, `RELATED`, `OBSTRUCTION`, or `UNRESOLVED`. The record preserves both projection identities, source-local names and meanings, unmatched assumptions, coordinate-level support/obstruction, and the exact reasons for refusing a merge. `ALIGNED` means only that the declared source-local coordinates agree for the proposed mapping purpose; formal equivalence remains Paper VI's responsibility.

A clean vocabulary-changing example is numerical bisection versus monotone threshold calibration. Their operation names differ, but under the narrow purpose "bounded monotone localization" both require an ordered domain, a bracketed monotone target, preserve a target-containing bracket, make progress by shrinking the admissible interval, terminate at a tolerance, and reconstruct a representative in the original target space. Paper III may therefore record a candidate alignment while preserving each donor's vocabulary and lineage.

A false-merge control uses the same-looking midpoint/discard-half surface sequence on a non-monotone response. The monotonicity/bracketing contract no longer holds and the invariant can fail, so the correct result is `OBSTRUCTION`; topology or lexical similarity cannot override that coordinate. A third control uses an opaque recovery procedure whose source does not establish progress or reconstruction semantics. Even when its known effects resemble a rollback method, the alignment remains `UNRESOLVED` rather than filling the missing coordinates from analogy.

Plural views are therefore first-class outcomes: structurally suggestive sources need not collapse into one global method. An obstruction certificate records exactly which coordinates prevent a merge and keeps both source projections recoverable.

3.10 Downstream structural-learning boundary

Paper III owns source-local method projection and candidate plural alignment. Paper VI owns claim-relative structural reduction, substitution and method-fibre membership. Paper IX may learn distributions over these versioned objects, and Paper X may later propose new method structures. Neither learned similarity nor generated structure changes the source projection or grants scientific authority. The bounded pilot reported in Section 5.7 tests only representation/alignment non-vacuity; it is not evidence of general extraction from arbitrary papers.

4 Public-reference mapping datasets

The empirical result in this paper uses two content-addressed *public-reference mapping datasets.* They contain already-structured scientific projections assembled from pinned external human/expert resources and deterministic standards. They are not newly commissioned expert gold and do not evaluate raw-text extraction.

Confirmatory case family	Cases
Different name / same referent	13
Polarity, modality, attribution, or context	13
Valid / invalid representation mapping	6
Total	**32**

Table 2: Case-family composition of the disjoint confirmatory public-reference holdout.

4.1 Initial public-reference set

The initial mapping study contains 32 cases generated from pinned revisions of MUSE, SciFact, and SciSchema. The portable gold archive has SHA-256 `35f9e39b75ff53b7f0ec82cd03ebcaaa82509ee0aea3f5b96aac` The evidence package binds the external-source revisions and evaluator identity in `evidence/public-reference-v` and retains a checksum manifest rather than redistributing restricted third-party text. This first result is treated as development/initial evidence; it is not pooled into the confirmatory verdict.

4.2 Disjoint confirmatory set

A second 32-case holdout was selected by an outcome-blind deterministic round-robin window over the sorted source pools, starting after the initial window and excluding all initial case identifiers. Its overlap with the first set is zero. Before confirmatory system outputs were evaluated, the execution manifest froze the portable gold hash `13a76c68c149c2552f3543babeca6e1ad5afe23c45ea9c0dc365c1445cf2782b` the MUSE/SciFact/SciSchema revisions, evaluator component identities, bootstrap seed, margins, and terminal rule.

The 13 different-name/same-referent cases come from the MUSE cross-domain stratum. The 13 polarity/modality/attribution/context cases come from a scientific-claim-verification stratum. The six representation-mapping cases span materials, physics, and psychology and use deterministic standards where appropriate. Each record retains its source authority class and exact frozen input identity.

4.3 What the public-reference labels mean

The evaluated decision is a mapping-level terminal over already-structured projections. Depending on the source relation and tested rule, the correct outcome can permit integration, forbid integration, or preserve an unresolved relation. The scoped study evaluates false merges and false splits under these frozen cases. It does not claim that a model recovered the semantic coordinates from raw papers.

The confirmatory set is particularly discriminating for the polarity/modality/attribution/context family: flat predicate canonicalization false-merges 6 of 13 cases (rate 0.4615), while ORION has zero false merges. The different-name/same-referent and representation-mapping strata are clean controls on this holdout: all evaluated rules are correct there. This localization is reported rather than hidden, because the observed pooled effect is not evidence that every coordinate is necessary.

4.4 Broader expert-atlas protocol is follow-up

The repository also preserves the prospective `P3.cross-domain-atlas.v1` protocol, annotation schema, handbook, adjudication policy, and 32 `seed-to-gold-v1` templates spanning a broader eight-family design. Those templates are not expert gold: no independent dual annotation, per-coordinate inter-annotator agreement, specialist adjudication, or prospectively bound raw-text model run exists for that broader object. It is therefore excluded from the scoped P3.C5 submission claim and remains `CANNOT_CHECK` follow-up science rather than being retrospectively presented as the dataset used here.

4.5 Data and licensing boundary

The public-reference package stores source identities, pinned revisions, locators, provenance and content commitments needed for replay. Third-party text is not vendored where redistribution is not permitted. The final submission archive will freeze the exact repository subject and stable public release URL for the legally shareable package; absence of a permanent archive at the current drafting stage does not change the scientific result.

5 Evaluation

The empirical claim in this paper is governed by the additive execution-frozen protocol `P3.public-reference-mapp` It was created after the initial 32-case result but before any confirmatory system outputs were examined. The predecessor result was allowed to motivate replication; it was not allowed to change the second holdout, evaluator, margins, or pass criteria. The broader `P3.cross-domain-atlas.v1` raw-text/expert study remains a separate unexecuted follow-up protocol.

5.1 Primary confirmatory hypothesis

The frozen replication hypothesis has two non-compensatory conditions:

False-merge superiority: the ORION-minus-flat false-merge point estimate must be at most -0.05, and the upper bound of the paired 95% bootstrap interval must be below zero.

False-split non-inferiority: the ORION-minus-exact-conservative false-split point estimate must be at most $+0.03$, and the upper bound of its paired 95% bootstrap interval must be at most $+0.03$.

The terminal is `PASS` only if both conditions hold. The confirmatory holdout contains 32 cases, has zero overlap with the initial 32-case set, and was bound by exact content hash before confirmatory outcome access.

5.2 Compared mapping rules

The confirmatory primary comparison uses three deterministic rules over the same already-structured mapping cases:

1. **ORION typed mapping calculus**: evaluates the scientific coordinates required by the case and can retain an obstruction rather than force a merge.
2. **Flat predicate canonicalization**: treats compatible surface predicates as sufficient merge evidence without the scoped coordinate distinctions. This is the predeclared false-merge comparator.

3. **Exact-coordinate conservative control**: requires exact coordinate compatibility and can abstain where it lacks such a match. This is the predeclared false-split/non-inferiority control.

These are mapping-layer comparators, not claims to reproduce full external KG, RAG, or schema-fusion systems. Stronger SCOPE/SCION-like and provenance/schema-contract baselines are useful post-outcome construct-validity pressure and are tracked under issue #280, but they cannot rewrite the frozen confirmatory status after the fact.

5.3 Secondary ablations

The frozen mapping evaluator also reports descriptive paired ablations. The important covered attacks are:

- **force compatibility without obstruction**: removes the explicit non-merge terminal and chooses a compatible mapping even where the scoped coordinates prohibit it;
- **remove modality/polarity/attribution/discourse**: removes the coordinate group carrying the main confirmatory discrimination;
- **remove referent**, **construct**, **measurement**, and **temporal context**: retained as zero-effect controls on this particular holdout.

The ablations are secondary/descriptive. The protocol explicitly forbids turning their intervals into new confirmatory significance claims or using zero effect on under-supported strata as evidence that a coordinate is unnecessary.

5.4 Metrics and statistics

The confirmatory decision uses case-level false-merge and false-split rates. Wilson 95% intervals describe individual rates. Paired differences use a 10,000-resample percentile bootstrap with frozen seed `20260817`. The mapping rules are deterministic, so the protocol uses one execution per case rather than artificial stochastic repeats. Results are reported pooled and descriptively by case family/source stratum. The initial 32 cases are never pooled into the confirmatory verdict.

The primary decision rule is therefore fixed entirely by the two paired comparisons above. No post-outcome margin change, exclusion, reweighting, or replacement baseline can change whether the confirmatory run passed.

5.5 Custody and fail-closed rules

Before confirmatory outputs, the manifest bound:

- portable confirmatory gold SHA-256 `13a76c68c149c2552f3543babeca6e1ad5afe23c45ea9c0dc365c1445cf2`
- zero overlap with the predecessor gold;
- exact MUSE, SciFact, and SciSchema source revisions;
- evaluator component Git identities;
- bootstrap seed and pass margins;

- a rule that any hash mismatch, source drift, evaluator drift, or overlap makes the result `CANNOT_CHECK` rather than silently re-evaluating a different study.

The gold is not an input to the evaluated mapping rule. The evaluator consumes the candidate mapping result and the protected expected decision after the system output has been sealed.

5.6 Reproducibility

The scoped result is reproduced by the deterministic public-reference scripts and immutable archives under `evidence/public-reference-v1.1-confirmatory/`. The exact confirmatory analysis, publication figures/tables, source registry, evaluator identities, and independent replay receipt are retained in the repository. The commands `make paper03-public-reference` and `make paper03-public-reference-publication` regenerate the scoped headline analysis and publication artifacts from committed inputs.

This reproducibility claim does not extend to the unexecuted raw-text expert atlas, full external RAG/schema baselines, generated-portrait recoverability, or downstream answer-quality evaluation. Those remain `CANNOT_CHECK` until their own prospective runs exist.

5.7 Bounded method-structure pilot

The present paper does *not* treat its public-reference scientific-meaning experiment as an evaluation of method-structure extraction. The larger expert-annotated method-structure corpus originally proposed for this extension—50–100 papers/algorithms across at least five fields with independent annotation—was not available for this submission track and therefore remains outside the empirical claim.

Instead, a prospectively frozen non-vacuity pilot uses six authored exact-ground-truth method pairs across numerical methods, graph algorithms and transactional workflows. The panel contains three structurally aligned pairs, two explicit obstructions and one source-insufficient `UNRESOLVED` case. Eight controlled corruptions delete or change an invariant, reconstruction map, dependency order, precondition, failure semantics, lineage, authority boundary, or an explicitly unknown progress coordinate. The typed representation/alignment path reproduces all six frozen pair labels and detects all eight material corruptions. A deliberately simple transcript/action-overlap comparator resolves only two of the six pair labels. The exact protocol, runner and result receipt are under `research/extensions/p1-p3-structure/`.

This result supports only a narrow claim: the typed source-local objects are discriminating on a small exact-ground-truth panel and can preserve an explicit unresolved state. It does not estimate annotation agreement on real papers, false merge/split rates over natural scientific prose, end-to-end extraction quality, or downstream P9 performance. Those coordinates remain `CANNOT_CHECK` until independently annotated evidence exists. The bounded terminal is therefore `P3_METHOD_PROJECTION_NARROWED`, not a general cross-domain method-learning result.

6 Results status

ORION Paper III has deliberately separate evidence layers. The scoped public-reference projection-to-mapping protocol has been executed twice: an initial frozen 32-case study followed by a disjoint, prospectively execution-frozen 32-case confirmation. The broader raw-text expert-atlas, generated-portrait recoverability, downstream utility, and full coordinate necessity questions are follow-up claims and remain `CANNOT_CHECK` rather than being treated as missing pieces of the reported mapping result. All numbers below come from immutable archived artifacts.

6.1 Initial public-reference mapping result

The public-reference builder used pinned external source authorities from MUSE, SciFact, and SciSchema and did not fill unsupported semantic coordinates with model guesses. The first deterministic build produced 32 already-structured mapping cases across five recorded strata and three case families: different-name/same-referent, polarity/modality/attribution/context, and valid/invalid representation mappings. The portable gold artifact has SHA-256 `35f9e39b75ff53b7f0ec82cd03ebcaaa82509ee0aea3f` A second freeze pass reproduced it byte-for-byte.

On those 32 cases, the ORION semantic comparison rule classified all expected relations correctly (accuracy 1.000), with false-merge rate 0.000 and false-split rate 0.000. Flat predicate canonicalization had accuracy 0.875 and false-merge rate 0.125, with no false splits. The paired ORION-minus-flat false-merge difference was -0.125 with a 95% paired percentile-bootstrap interval of $[-0.250, -0.03125]$ (10,000 resamples; seed 20260817).

An exact-coordinate conservative control also produced zero false merges but abstained on 0.125 of cases and achieved accuracy 0.875. The initial result therefore motivated a prospective replication but did not itself become the final confirmatory authority.

6.2 Prospectively frozen confirmatory replication

A second, disjoint 32-case holdout was selected with the same outcome-blind round-robin construction at selection offset 32, excluding every initial case identifier. It was frozen before any confirmatory ORION or control output was examined. Its portable SHA-256 is `13a76c68c149c2552f3543babeca6e1ad5afe` The execution manifest `P3.public-reference-mapping.v1.1-confirmatory` bound that hash, upstream source revisions, evaluator Git identities, bootstrap seed, margins, and the terminal rule before evaluation.

The predeclared rule required both (i) an ORION-minus-flat false-merge difference at most -0.05 with the upper end of its paired 95% bootstrap interval below zero, and (ii) an ORION-minus-exact false-split difference no larger than $+0.03$ with the interval upper bound no larger than $+0.03$. The holdout passed both conditions. ORION had false-merge rate 0.000 and false-split rate 0.000. Flat predicate canonicalization had false-merge rate 0.1875 and accuracy 0.8125. The paired ORION-minus-flat false-merge difference was -0.1875 with 95% interval $[-0.34375, -0.0625]$. The ORION-minus-exact false-split difference was 0.000 with interval $[0.000, 0.000]$. The confirmatory primary verdict was therefore **PASS** under the rule frozen before those outputs.

The effect is localized to the strata that discriminate the tested rules. Among 13 polarity/modality/attribution/context cases, flat canonicalization false-merged 6/13 (0.4615), whereas ORION false-merged none. The 13 different-name/same-referent cases and six representation-mapping cases were correct under all compared rules. These strata are reported descriptively and are not pooled with the initial 32 cases to inflate confirmatory sample size.

6.3 Covered ablations

On the confirmatory holdout, forcing compatibility without the explicit obstruction distinction increased false merges by $+0.1875$ (95% paired bootstrap interval $[+0.0625, +0.34375]$) and reduced accuracy by the same absolute amount. Removing the modality/polarity/attribution/discourse coordinate group produced the same false-merge increase on the covered claim-verification cases.

Removing referent, construct, measurement, or temporal context has zero measured effect on this holdout. Those zero effects are preserved as coverage limitations, not interpreted as proof that the coordinates are unnecessary or dispensable. The confirmatory protocol treats all ablations as descriptive secondary analyses rather than new significance claims.

6.4 Independent verification and local falsifiers

A separate implementation independently reproduced the P3.C5 confirmatory mapping result and its case-level decisions under the programme verification stack. The verification is bounded to the public-reference mapping claim; it does not promote the raw-text or downstream follow-up claims.

Synthetic exact-world tests separately exercise referent and construct identity, measurement equivalence, context, GLUE, OBSTRUCTION, PLURAL_VIEW, preservation conditions, and typed mapping hypotheses. These remain engineering/semantic falsifiers rather than external empirical evidence.

6.5 Explicit follow-up nonclaims

The scoped result does not establish:

- **Raw-text end-to-end superiority (P3.C7):** no prospectively bound raw-text extractor/model comparison against strong RAG/schema systems supports this paper's headline;
- **Generated-portrait recoverability or downstream utility (P3.C8):** no frozen reader/downstream task result is promoted here;
- **Universal coordinate necessity (P3.C6):** targeted cases for referent, construct, measurement, temporal context and additional obstruction families belong to follow-up issue #280;
- **Executed expert eight-family gold:** the repository's broader seed templates and annotation protocol remain prospective rather than being relabelled as the public-reference dataset.

Those nonclaims remain visible so the replicated mapping result cannot be read as authority for the broader research programme. Any future positive must be added through a new prospectively frozen artifact rather than by widening the current claim in prose.

7 Limitations, ethics, and reproducibility

7.1 Limitations

Scope of the reported external result. The reported external result is the public-reference projection-to-mapping study, not the original end-to-end expert atlas. It contains two separately frozen 32-case mapping sets assembled from pinned public human/expert resources; the confirmatory set is disjoint at the case level from the initial set. These cases are already structured scientific projections. They therefore test mapping and obstruction semantics, not whether a model can reliably extract all required semantic coordinates from raw papers.

Expert subjectivity and theory-ladenness. Scientific identity and measurement equivalence judgments are theory-laden and may depend on disciplinary background. The broader `P3.cross-domain-atlas.v1` protocol mitigates this through a written handbook, independent double annotation, coordinate-wise agreement reporting, domain-expert escalation, and an UNRESOLVED state. That broader expert-gold study has not been executed, so this manuscript does not report inter-annotator agreement or present seed templates as expert gold.

Domain and case-family coverage. The public-reference route covers only the case families supported by its pinned source authorities. Its confirmatory effect is concentrated in polarity/modality/attribution/context cases; different-name/same-referent and representation-mapping cases are useful controls but do not discriminate the compared rules on that holdout. The result therefore cannot establish universal cross-domain adequacy or necessity of every semantic coordinate. In particular, zero measured ablation effects for referent, construct, measurement, or context on under-supported strata are coverage limitations, not evidence that those coordinates are dispensable.

Evolving terminology. Scientific terminology and operationalization change over time. The current mapping study is bound to specific source revisions and does not establish robustness to temporal drift, historical conceptual change, or cross-lingual terminology.

Ontology and schema dependence. The semantic-coordinate representation is a design choice. Other representations may encode additional causal, probabilistic, temporal, or normative structure. The current evidence supports a bounded typed mapping/obstruction result; it does not establish that the chosen coordinate system is uniquely correct or optimal.

Extraction error propagation. The text-to-scientific-meaning extraction layer remains an unexecuted part of the broader protocol. Extraction errors could propagate into mapping, GLUE, and obstruction decisions. Because no prospectively bound raw-text extractor run supports the headline result, this paper does not report stage-attributed extraction error or claim end-to-end superiority over raw-text/RAG/schema systems.

Recoverability and downstream utility remain open. The framework is designed so global statements retain source projections, mapping paths, preservation conditions, and non-preserved coordinates. Local tests verify those mechanics, but the current external study does not establish reader-level recoverability of generated portraits or downstream scientific-answer improvement. Those claims remain `CANNOT_CHECK` pending their own frozen evaluations.

7.2 Ethics

The scoped public-reference study reuses published research resources and does not collect new human-participant data. The broader expert-annotation protocol is prospective and has not been executed. ORION's outputs, including OBSTRUCTION and PLURAL_VIEW, should be presented as provenance-bound records of incompatibility or unresolved mapping rather than as judgments that one scientific perspective is intrinsically superior. Source lineage is retained so a downstream reader can inspect the evidence and conditions behind an integration decision.

7.3 Reproducibility

The original end-to-end design remains frozen in `protocol/PROTOCOL_V1.json` without pretending its unbound expert-gold/model identities have executed. The scoped public-reference route is separately content-addressed: the initial portable gold and its disjoint confirmatory holdout are hash-bound, and the confirmatory execution manifest freezes the holdout hash, upstream source revisions, evaluator code identities, bootstrap seed, practical margins, and decision rule before confirmatory outputs. The deterministic repository commands regenerate the public-reference metrics and publication artifacts from those immutable inputs. Broader expert-gold, raw-text, recoverability, and downstream-utility results require new prospective artifacts rather than reinterpretation of the current mapping evidence.

8 Conclusion

ORION Paper III asks whether a global knowledge portrait — an integration of scientific claims across disciplines and measurement systems — can be built without treating surface similarity as scientific identity. The framework makes three structural commitments.

First, *projection before mapping.* Each source retains a local projection whose scientific meaning is represented by typed coordinates and bound to exact provenance before any cross-source decision is made. This keeps the question of what a source means separate from the question of which sources may be combined.

Second, *GLUE-or-obstruction.* Integration is decided over typed mapping hypotheses: projections are merged only when the relevant compatibility conditions are satisfied; otherwise the system reports a typed obstruction, contextual difference, or plural portrait rather than silently flattening the distinction.

Third, *recoverability.* The intended global portrait exposes its construction — source projections, mapping path, preservation conditions, and non-preserved coordinates — so that a downstream consumer can audit a global statement back to its origins.

The formal projection/mapping semantics are verified by local exact worlds. The prospectively frozen public-reference route provides a bounded external check of the mapping layer without requiring a newly commissioned annotation team. On the disjoint 32-case confirmatory holdout assembled from pinned public human/expert resources, ORION's typed comparison produced no false merges, whereas flat predicate canonicalization false-merged 0.1875 of cases; the paired difference was -0.1875 with a 95% bootstrap interval of $[-0.34375, -0.0625]$. Against the exact-coordinate conservative control, the false-split difference was 0.000 with interval $[0.000, 0.000]$, so the predeclared confirmatory rule passed. Forcing compatibility without the obstruction distinction increased false merges by $+0.1875$ with interval $[+0.0625, +0.34375]$.

That result is evidence for the mapping calculus, not for the full envisioned research system. The stronger questions remain open: whether a prospectively bound extractor can recover the required scientific projections from raw papers, whether the same advantage survives the full eight-family cross-domain study against resource-matched synthesis/schema baselines, whether generated global statements are practically recoverable, and whether obstruction improves downstream scientific decisions. Those claims remain `CANNOT_CHECK` until their own frozen evidence exists.

The narrower result nevertheless falsifies one tempting assumption on externally grounded cases: equal or normalized predicates are not sufficient license to merge scientific representations. The broader ORION proposal remains a follow-up scientific programme whose claims must be earned separately rather than retroactively attached to the replicated mapping result.

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